6 (a) (i) Effective resistance of Q and LDR,

$$R_{eff} = \left(\frac{1}{R_Q} + \frac{1}{R_{LDR}}\right)^{-1}$$

$$= \left(\frac{1}{6.0} + \frac{1}{8.0}\right)^{-1}$$

$$= 3.4286 \text{ k}\Omega$$

$$I_{A_1} = \frac{E}{R_T}$$

$$= \frac{9.0}{(3.4286 + 4.0) \times 10^3}$$

$$= 1.2115 \times 10^{-3} = 1.21 \times 10^{-3} \text{ A}$$

(ii) Potential difference across Q and LDR,

$$V_{eff} = \frac{R_{eff}}{R_{eff} + R_P} \times E$$

$$= \left(\frac{3.4286}{3.4286 + 4.0}\right) \times 9.0$$

$$= 4.1539 \text{ V}$$

$$I_{A_2} = \frac{V_{LDR}}{R_{LDR}}$$

$$= \frac{4.1539}{8.0 \times 10^3}$$

$$= 5.1924 \times 10^{-4} = 5.19 \times 10^{-4} \text{ A}$$

(b) (i) Resistance of the LDR increases when light intensity is lowered. The effective resistance of Q and the LDR increases.

Since the potential difference across P and Q is the same at 9.0 V, by the potential divider principle, the potential difference across Q is a larger proportion of the 9.0 V. Hence the potential difference across Q increases.

OR

Resistance of the LDR increases when light intensity is lowered. The overall resistance of the circuit increases.

Current from the battery decreases. As the resistance of P is the same, the potential difference across P decreases. Since the potential difference across P and Q remains the same at 9.0 V, this means the potential difference across Q increases.

(ii) Since the potential difference across Q increases, the current through Q increases for the same resistance of Q.

Total current in the circuit is the sum of the current through Q and the LDR. Since the total current from the battery decreases due to overall increase in resistance, and the current in Q increases, this means the current through the LDR decreases. Hence the current reading on ammeter A_2 decreases.

7 **(c)** In a nuclear fusion reaction, two low nucleon number (OR lighter) nuclei combine into a